Numerical Experiments on Quantum Relaxation

CHEM 5260 Midterm Project

Ubaidullah Hassan | March 2026

1. Project setup

The two Hamiltonians studied here are

$$H_A = E_1|1\rangle\langle 1| + \sum_{k=1}^{N} E_k|k\rangle\langle k| + \sum_{k=1}^{N} V_{1k}\big(|1\rangle\langle k| + |k\rangle\langle 1|\big), \tag{1}$$

$$H_B = E_0|0\rangle\langle 0| + E_1|1\rangle\langle 1| + V_{01}\big(|0\rangle\langle 1| + |1\rangle\langle 0|\big) + \sum_{k=1}^{N} E_k|k\rangle\langle k| \tag{2}$$

$$+ \sum_{k=1}^{N} V_{1k}\big(|1\rangle\langle k| + |k\rangle\langle 1|\big). \tag{3}$$

The initial states are $|1\rangle$ for model A and $|0\rangle$ for model B. Following the project prompt, the continuum energies were taken uniformly over a window of span ΔE centered on E_1, so the mean spacing is

$$\epsilon = \frac{\Delta E}{N}, \qquad \rho = \frac{1}{\epsilon} = \frac{N}{\Delta E}.$$

Without loss of generality I set $E_1 = 0$ and used units with $\hbar = 1$, so time is measured in inverse-energy units.

The populations of interest are

$$P_1(t) = |\langle 1|\psi(t)\rangle|^2 \quad \text{(model A)}, \qquad P_0(t),\ P_1(t),\ P_k(t) = \sum_k |\langle k|\psi(t)\rangle|^2 \quad \text{(model B)}.$$

2. Numerical method

Each Hamiltonian was diagonalized exactly,

$$H = UDU^\dagger, \qquad |\psi(t)\rangle = Ue^{-iDt}U^\dagger|\psi(0)\rangle,$$

and all populations were computed directly from the resulting state vector. This gives the full unitary dynamics with no perturbative approximation. For comparison with the wide-band theory discussed in class, the Golden-Rule width

$$\Gamma = 2\pi\rho V^2$$

was used whenever all couplings had equal magnitude $|V_{1k}| = V$.

Two additional time scales are useful for interpreting the numerics:

$$\tau_{\text{mem}} \sim \Delta E^{-1} \quad \text{(continuum memory time)}, \qquad T_{\text{rec}} \sim \frac{2\pi}{\epsilon} \quad \text{(finite-size recurrence scale)}.$$

A broad, quasi-continuous band is therefore expected to show exponential relaxation only when

$$\epsilon \ll \Gamma \ll \Delta E.$$

This criterion organizes all of the results below.

Table 1: Parameter sets used in the figures.

Figure	Model	N	ΔE	V_{1k}	V_{01}	$E_0 - E_1$	Purpose
1	A	200	8	0.05	—	—	Reference case in the wide-band regime
2a	A	40, 200	8	0.112, 0.05	—	—	Finite-size comparison at matched initial width
2b	A	200	1, 8	0.05	—	—	Breakdown when the band is no longer broad
3	B	200	8	0.05	0.06, 0.25	0	Overdamped vs. underdamped resonant transfer
4	B	200	8	0.05	0.25	0, 0.5, 1.0	Effect of detuning

3. A useful structural observation: random signs do *not* change the answer

If

$$V_{1k} = s_k V, \qquad s_k = \pm 1,$$

then model A with random signs is related to model A with all-positive couplings by the unitary basis change

$$U_A = |1\rangle\langle 1| + \sum_k s_k |k\rangle\langle k|, \qquad H_{\{s_k\}} = U_A H_+ U_A^\dagger.$$

Exactly the same statement holds for model B with

$$U_B = |0\rangle\langle 0| + |1\rangle\langle 1| + \sum_k s_k |k\rangle\langle k|.$$

Since the initial states $|1\rangle$ (model A) and $|0\rangle$ (model B) are unchanged by these transformations, all physical populations are identical. Therefore:

- random *signs* by themselves do not produce dephasing or disorder effects here;
- only changes in the *magnitudes* $|V_{1k}|$ or in the energy distribution E_k can modify the dynamics.

For that reason, the figures below focus on the parameters: N, ΔE, V_{1k}, V_{01}, and $E_0 - E_1$.

4. Results for model A

4.1 Reference case: exponential decay after a short transient

Figure 1 shows a representative case with $N = 200$, $\Delta E = 8$, and $V = 0.05$. The corresponding spacing is $\epsilon = 0.04$ and the Golden-Rule width is

$$\Gamma = 2\pi \frac{N}{\Delta E} V^2 \approx 0.393.$$

The numerical survival probability is nearly a straight line on the semilog plot after a very short initial transient, which is the expected signature of exponential decay. The deviation at very early times is also expected: unitary dynamics always begins quadratically in time, not exponentially.

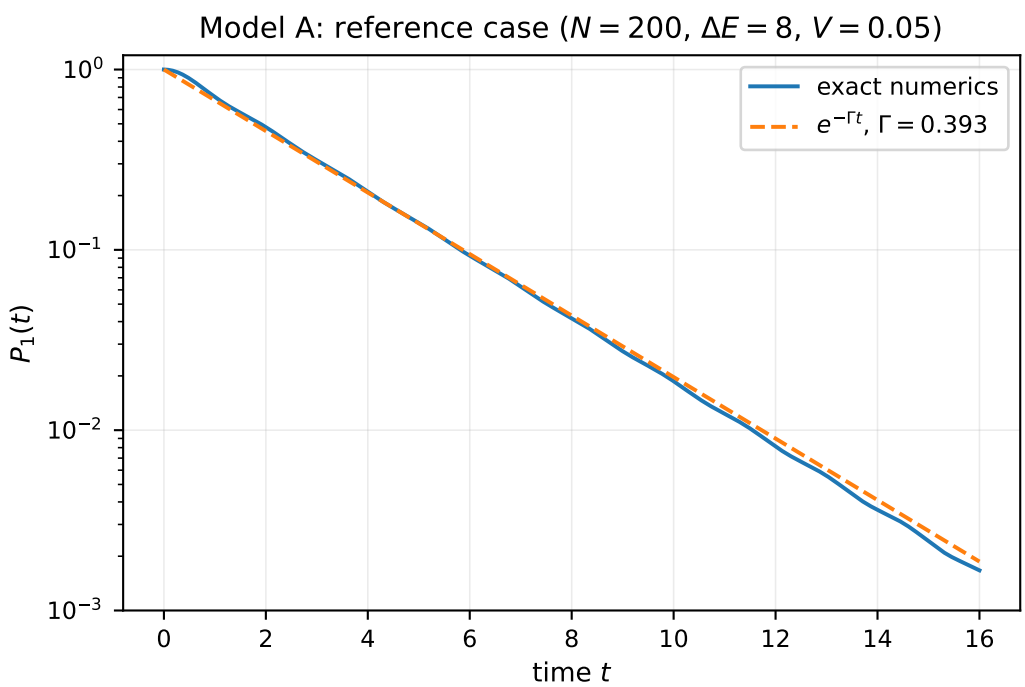

Figure 1: Model A in the broad-band regime. After the initial quadratic transient, the exact numerical result follows the Golden-Rule envelope $e^{-\Gamma t}$ closely.

4.2 When exponential relaxation fails

Figure 2 shows the two main ways in which the exponential law breaks down.

Finite-size breakdown. In panel (a), the early decay is matched, but the smaller continuum ($N = 40$) develops a large revival near $t \sim 35$. This is the signature of a discrete spectrum: amplitude that leaves $|1\rangle$ can rephase and return.

Narrow-band breakdown. In panel (b), ΔE is reduced from 8 to 1 while V is kept fixed. Since $\Gamma \propto 1/\Delta E$, the band is no longer broad compared with the relaxation width, and the survival probability becomes strongly oscillatory rather than monotonic.

Summary for model A. The numerics support the standard picture from class very clearly:

- if $\epsilon \ll \Gamma \ll \Delta E$, the doorway state relaxes approximately exponentially;

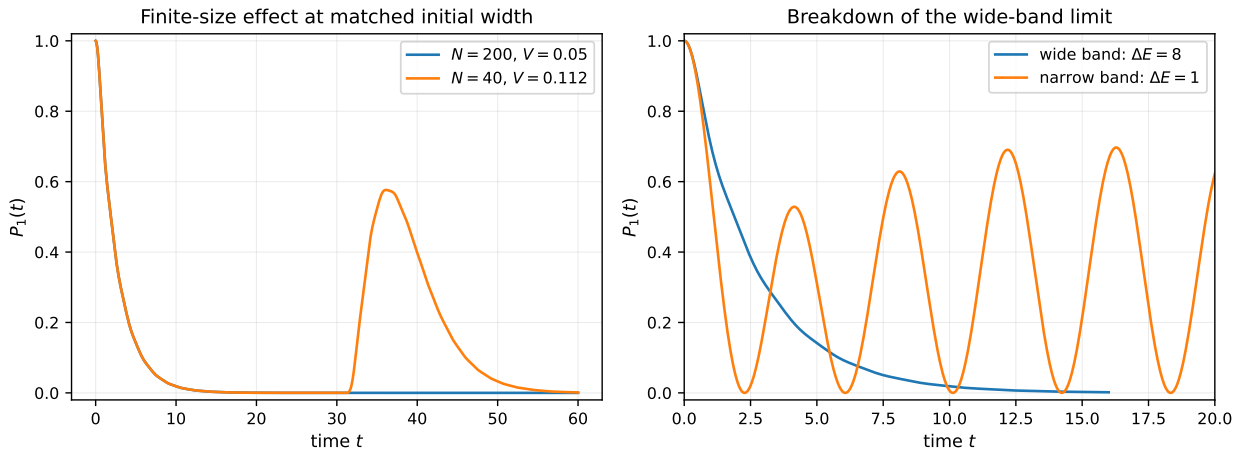

Figure 2: Two distinct non-Markovian effects in model A. Left: decreasing N makes the continuum visibly discrete and produces a strong recurrence. Right: decreasing ΔE invalidates the wide-band approximation and generates large oscillations.

- if N is too small, revivals appear on the scale $T_{\text{rec}} \sim 2\pi/\epsilon$;
- if ΔE is too small, the continuum has memory and the decay is no longer well described by a single rate.

5. Results for model B

In model B, state $|1\rangle$ is still the doorway to the continuum, but population must first flow from $|0\rangle$ to $|1\rangle$ through V_{01}. In the same wide-band limit used above, the continuum acts approximately as an imaginary self-energy on $|1\rangle$,

$$H_{\text{eff}} \approx \begin{pmatrix} E_0 & V_{01} \\ V_{01} & E_1 - i\Gamma/2 \end{pmatrix},$$

so the problem becomes a damped two-level system.

5.1 Resonance: overdamped and underdamped transfer

Figure 3 keeps $E_0 = E_1 = 0$ and compares two values of V_{01} at fixed continuum width $\Gamma \approx 0.393$.

For $V_{01} = 0.06$, transfer from $|0\rangle$ is slow and nearly monotonic: population leaks through $|1\rangle$ into the continuum without visible oscillations. For $V_{01} = 0.25$, the coherent $0 \leftrightarrow 1$ exchange is strong enough to survive against the damping, so damped oscillations appear in both $P_0(t)$ and $P_1(t)$ before the population ends up in the manifold states.

5.2 Detuning suppresses relaxation

Figure 4 fixes $V_{01} = 0.25$ and increases the detuning $E_0 - E_1$. As the detuning grows,

- the maximum population reaching the doorway state $|1\rangle$ decreases,

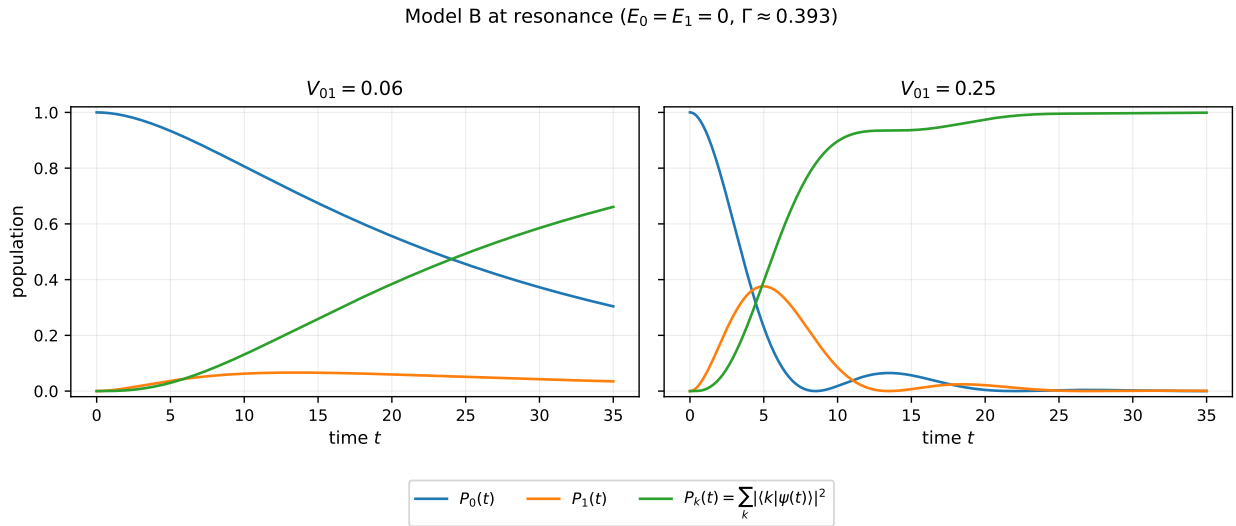

Figure 3: Model B at resonance. Small V_{01} gives overdamped transfer; larger V_{01} gives damped oscillatory transfer before the probability settles into the continuum.

- the survival probability $P_0(t)$ stays larger for longer times,
- the total transferred population $P_k(t)$ grows more slowly and reaches a smaller value on the plotted time interval.

This is the expected off-resonant suppression of doorway-mediated relaxation.

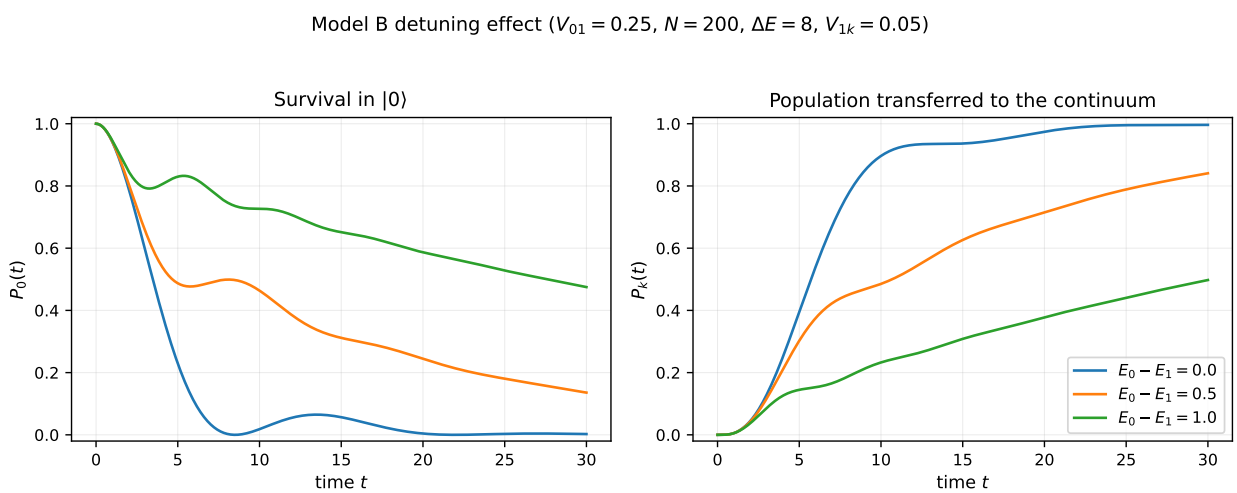

Figure 4: Model B with fixed $V_{01} = 0.25$. Detuning weakens the coupling between $|0\rangle$ and the decaying doorway state $|1\rangle$, so relaxation into the continuum becomes progressively less efficient.

6. Conclusions

The numerical experiments can be summarized by

1. **Model A reproduces exponential relaxation only in the broad-band, quasi-continuum limit.** The practical criterion is $\epsilon \ll \Gamma \ll \Delta E$.
2. **Finite N produces revivals.** Even when the short-time decay looks exponential, a sparse continuum eventually reveals its discreteness through recurrences.
3. **A narrow band produces memory effects.** Once the relaxation width is no longer small compared with the band span, the decay ceases to be Markovian and becomes oscillatory.
4. **Model B is a damped doorway problem.** State $|1\rangle$ mediates irreversible transfer from $|0\rangle$ into the continuum. The ratio V_{01}/Γ controls whether that transfer is overdamped or oscillatory.
5. **Random signs of V_{1k} are irrelevant in these two models.** They can be removed exactly by a basis transformation, so they do not change any population dynamics.

Overall, the computations support that exponential relaxation is not a generic consequence of coupling to many states; it emerges only when the continuum is sufficiently dense and sufficiently broad.